UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 001 · Introduction to second and higher order DE with constant coefficients · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·12	Unit 2 deck 001, slide 12 (rendered .pptx)
T001 00:17:38	Unit 2 AV-summary transcript 001 (“Introduction to higher order differential equations, complementary function and particular integral”), segment start
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s3 3d	Past paper: UE23MA141A Jan 2024 set 3, question 3.d

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A Formula sheet

1 Linear differential equations and their order

E1.1 · First-order linear DE

ESA ★

$$\frac{dy}{dx} + P\,y = Q$$

GIVES The general linear equation of order 1.

APPLIES P, Q functions of x alone.

NOTE Solved in the earlier unit by an integrating factor; not part of Unit 2.

S001·6 · S001·7 · T001 00:07:04

E1.2 · Second-order linear DE

ESA ★★

$$\frac{d^2y}{dx^2} + P_1\frac{dy}{dx} + P_2\,y = X$$

GIVES The general linear equation of order 2.

APPLIES P_1, P_2, X functions of x alone.

NOTE The lecture writes $P_0y'' + P_1y' + P_2y = X$ with $P_0 \neq 0$; dividing by P_0 gives the slide's form.

S001·7 · T001 00:07:22

E1.3 · Linear DE of order n

ESA ★★

$$\frac{d^ny}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + a_2\frac{d^{n-2}y}{dx^{n-2}} + \cdots + a_ny = X$$

GIVES The general linear equation of order n.

APPLIES $a_1, \ldots, a_n$ functions of x or constants; X a function of x alone.

NOTE Slide 001·7 writes $P_1, \ldots, P_n$; slide 001·8 onward $a_1, \ldots, a_n$.

S001·7 · S001·8 · T001 00:08:38 · T001 00:09:46

2 Constant or variable coefficients; homogeneous or not

E2.1 · Homogeneous and non-homogeneous ESA ★★

$$X = 0 : \text{ homogeneous}; \qquad X \not\equiv 0 : \text{ non-homogeneous}$$

GIVES The type of the equation from its right-hand side.

APPLIES Any LDE (the deck states it for constant coefficients).

NOTE Slide 001·10: $2y'' - 3y' + 9y = 0$ (homogeneous) and $2y'' - 3y' + 9y = x\sinh x$ (non-homogeneous).

S001·10 · T001 00:13:22 · T001 00:13:52

3 The differential operator D and $f(D)$

E3.1 · Operator notation ESA ★★★

$$D = \frac{d}{dx}, \quad D^2 = \frac{d^2}{dx^2}, \quad D^3 = \frac{d^3}{dx^3}, \ \ldots, \ D^n = \frac{d^n}{dx^n}$$

GIVES Shorthand for derivatives: $Dy = y'$, $D^2y = y''$.

APPLIES Always; ESA questions use it.

NOTE D^2y means $\frac{d^2y}{dx^2}$, not $\left(\frac{dy}{dx}\right)^2$.

S001·11 · T001 00:15:16

E3.2 · The equation as $f(D)y = X$ ESA ★★★

$$f(D)\,y = X, \qquad f(D) = D^n + a_1D^{n-1} + a_2D^{n-2} + \cdots + a_n$$

GIVES The compact form used in every later class.

APPLIES Constant a_i: $f(D)$ is a polynomial in D.

NOTE a_n multiplies y itself ($D^0 = 1$).

S001·11 · S001·12 · T001 00:16:29

E3.3 · Factorising $f(D)$ (constant coefficients) ESA ★★

$$(D-\alpha)(D-\beta)y = (D-\beta)(D-\alpha)y = \big[D^2 - (\alpha+\beta)D + \alpha\beta\big]y$$

GIVES $f(D)$ may be factorised into linear factors, taken in any order.

APPLIES α, β constants.

NOTE "It obeys the laws of algebra" (T001 00:16:49). Fails for variable coefficients.

T001 00:16:49 · T001 00:17:07

E3.4 · Preview: $f(D)$ acting on e^{mx} ESA ★★★

$$D^k e^{mx} = m^k e^{mx} \implies f(D)e^{mx} = f(m)e^{mx}$$

GIVES e^{mx} solves $f(D)y = 0$ exactly when $f(m) = 0$, the auxiliary equation.

APPLIES Constant coefficients.

NOTE Not in this deck; the auxiliary equation and all CF cases are class 002.

ADDED FOR CLARITY S001·16

4 Linear independence and the complementary function

E4.1 · Linear dependence and independence ESA ★

$$\text{independent:}\quad c_1 f_1 + c_2 f_2 + \cdots + c_n f_n = 0 \;\Rightarrow\; c_1 = c_2 = \cdots = c_n = 0$$

GIVES The test for a set of functions.

APPLIES On an interval I; "$= 0$" means for every x in I.

NOTE Dependent: some constants not all zero make the combination vanish.

S001·13 · T001 00:19:05 · T001 00:20:17

E4.2 · Two functions ESA ★★

$$y_1,\ y_2 \text{ independent} \iff \frac{y_1}{y_2} \neq \text{constant}$$

GIVES A quick test for a pair.

APPLIES Two functions only.

NOTE Neither is a constant multiple of the other (T001 00:22:00).

S001·14 · T001 00:21:01 · T001 00:22:00

E4.3 · Wronskian of two functions ESA ★

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

GIVES $W \neq 0$ (at some point of I) shows y_1, y_2 independent; for two solutions of the same LDE, $W \equiv 0$ means dependent.

APPLIES For n functions: the $n \times n$ determinant of the functions and their first $n - 1$ derivatives.

NOTE Not in the deck. $W(e^x, e^{2x}) = e^{3x}$, $W(e^x, 3e^x) = 0$. Used in class 011 (variation of parameters).

ADDED FOR CLARITY S001·14

E4.4 · Superposition: the complementary function ESA ★★★

$$y_c = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

GIVES The complete solution of the homogeneous LDE $f(D)y = 0$ of order n.

APPLIES $y_1, \ldots, y_n$ are n linearly independent solutions.

NOTE Second order: $y_c = c_1 y_1 + c_2 y_2$. Exactly n arbitrary constants.

S001·15 · S001·16 · T001 00:24:29 · T001 00:25:14

5 The general solution: $y = \text{CF} + \text{PI}$

E5.1 · Complete solution of $f(D)y = X$ ESA ★★★

$$y = y_c + y_p = \text{CF} + \text{PI}$$

GIVES The answer to every non-homogeneous LDE.

APPLIES y_c: complete solution of $f(D)y = 0$; y_p: any one solution of $f(D)y = X$.

NOTE All 24 Unit 2 ESA occurrences are answered in this form.

S001·12 · S001·17 · T001 00:27:16 · T001 00:27:34

E5.2 · Why it works (linearity) ESA ★★

$$f(D)(y_c + y_p) = f(D)y_c + f(D)y_p = 0 + X = X$$

GIVES $y_c + y_p$ solves $f(D)y = X$ and has all n constants.

APPLIES $f(D)$ linear.

NOTE Also: the PI of $X_1 + X_2$ is the sum of the PIs of X_1 and X_2 (W5.3).

ADDED FOR CLARITY S001·17

B Symbol glossary

Symbol	Name	Meaning	Appears in
x	Independent variable	The variable we differentiate with respect to	all
y	Dependent variable	The unknown function $y(x)$	all
X	Right-hand side	A function of x alone; $X = 0$ homogeneous. Not the variable x	E1.2, E1.3, E3.2
n	Order	Highest derivative; also the number of constants in the CF	E1.3, E4.4
$P,\ Q$	First-order coefficients	Functions of x in $y' + Py = Q$	E1.1
$P_1, \ldots, P_n$	Coefficients (slide 001·7)	Functions of x; the lecture adds $P_0 \neq 0$	E1.2
$a_1, \ldots, a_n$	Coefficients (slides 001·8 on)	Functions of x or constants	E1.3, E3.2
$k_1,\ k_2$	Constant coefficients	In $y'' + k_1 y' + k_2 y = 0$ (slide 001·15)	S001·15
D	Differential operator	$\frac{d}{dx}$; $D^n = \frac{d^n}{dx^n}$	E3.1
$f(D)$	Linear differential operator	$D^n + a_1 D^{n-1} + \cdots + a_n$	E3.2, E5.1
$f_1, \ldots, f_n$	Functions	Tested for linear dependence	E4.1
$c_1, \ldots, c_n$	Constants	In a linear combination; the arbitrary constants of the CF	E4.1, E4.4
$y_1, \ldots, y_n$	Independent solutions	Of the homogeneous LDE	E4.4
W	Wronskian	$y_1 y_2' - y_2 y_1'$ (added for clarity)	E4.3
y_c	Complementary function	Complete solution of $f(D)y = 0$	E4.4, E5.1
y_p	Particular integral	One solution of $f(D)y = X$	E5.1
m	Root of the auxiliary equation	$f(m) = 0$ (preview of class 002)	E3.4

Remark. The slides use three letters for coefficients: P, Q for first order, $P_1, \ldots, P_n$ in slide 001·7 and $a_1, \ldots, a_n$ from slide 001·8 on; k_1, k_2 for the second-order homogeneous equation (slide 001·15). The lecture also writes a leading coefficient P_0 or $a_0 \neq 0$; the slides divide it out. X (capital) is the right-hand side function of x, not the variable x.